

Decision-making on crop sequence diversification in the Argentine pampas

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ABSTRACT

Context: Heterogeneity in land use decisions is widespread in the Argentine Pampas. Given the prevailing simplified cropping systems, a move towards more sustainable intensification implies enhancing crop rotation diversification. Despite the regional and global significance of agriculture in the Pampas, there is limited information regarding the determinants influencing land use decisions.

Objective: This study analyses the factors that influence the decision to diversify crop sequences in Rolling Pampa, the corn belt of Argentina.

Methods: From April 2020 to March 2021, a survey was conducted in the Arrecifes river basin, located in northern Buenos Aires province, Argentina. The survey involved 120 farmers who were chosen using a stratified sampling technique. The survey collected information on land use, specifically regarding crop sequences in 2017–2018, 2018–2019, and 2019–2020 cropping seasons, as well as the structural variables of farms included size, tenure, and location. The survey also gathered data on farmers' characteristics, the land use decision-making process, and objectives and constraints in shaping land use. Employing a probit statistical model and conducting independence tests, the study analyzed factors influencing crop sequences selection.

Results and Conclusions: The findings indicate a positive correlation between land use diversification and farm size, land tenure, and the inclusion of livestock activities. Additionally, regional differences in the degree of diversification were noted. A negative correlation emerged between diversification and the years of experience of decision-makers. The study also found that the likelihood of diversification is significantly affected by the objectives related to reducing economic risk and maintaining high levels of soil organic matter, as well as the limitation of land availability for cultivation.

Significance: To promote suitable paths towards sustainable intensification of agriculture in the Argentine Pampas, it is essential to identify the factors that influence farmers' land use decisions. This study contributes to the understanding of the decision-making processes regarding crop rotations and cropping system diversification in the region, providing valuable insights for policymakers, advisors, and agricultural stakeholders to design targeted strategies for sustainable agricultural development.

1. Introduction

Land use heterogeneity in the Argentine Pampas ranges from simplified crop sequences, which mostly include a single summer crop, to more diverse and intensive crop sequences, with high proportions of double cropping within the same growing season (Pacín and Oesterheld, 2014; Cabrini et al., 2019). The simplification of the sequences generally stems from the prevalence of soybean in grain crop production systems. This is primarily due to lower costs per unit area and higher economic returns on invested capital compared to other alternatives. (Lanteri,

2009; Pengue, 2009; Cabrini and Calcaterra, 2016).

Widespread use of simplified sequences has raised environmental concerns. These concerns include reduced soil productivity (Novelli et al., 2011), biodiversity loss (Beck et al., 2005), and nutrient imbalances (Novelli et al., 2023), particularly related to phosphorus and nitrogen (Jobbágy et al., 2021). Baumgärtner and Quaas (2010) further highlight that reduced agrobiodiversity impairs the ecosystem's natural insurance capacity against environmental risks, which becomes increasingly important in the context of climate change. In response to such challenges, the sustainable intensification proposition entails

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embracing more diverse and intensive crop sequences (Caviglia et al., 2004; Caviglia and Andrade, 2010; Andrade et al., 2015, 2017; Cano et al., 2023).

To select and disseminate suitable sustainable intensification pathways for the Pampas, it is essential not only to better understand the productive and environmental aspects of different crop sequences, but also to recognize the factors that influence land use decision-making. Furthermore, in a political context where voluntary action is preferred over government regulation, it is crucial to deepen our understanding of decision-making and enhance the prediction of farmers' behavior (Dessart et al., 2019; Thomas et al., 2019; Streletskaia et al., 2020; Sok et al., 2021).

When analyzing the variability in decisions made by farmers, the multi-attribute utility theory is an appropriate approach (Gómez-Limón et al., 2003; Berkhout et al., 2011), as it allows for the consideration of the various objectives pursued by farmers and the importance they assign to each of them. The basic assumption upon which this theory is based is that the decision-maker selects the alternative that yields the highest utility after considering multiple attributes in a set of possible alternatives.

Several objectives have been identified as determinants of farmers' decision-making, such as maximizing economic short-term expected outcomes, reducing risk, avoiding environmental harm, preserving long-term resource productivity, ensuring business continuity, managing off-farm income opportunities (Ochoa et al., 2019; Murphy, 2001; Pfeiffer et al., 2009) and simplifying rural tasks (Sattler and Nagel, 2010; Berkhout et al., 2011; Jansen, 2011; Mandryk et al., 2014; Arora et al., 2015). Often, these objectives conflict with each other, making the decision-making process more complex.

In turn, the attitudes towards diversification are influenced by different characteristics of decision makers and their farms. Studies conducted in different parts of the world have shown a relationship between crops diversification and characteristics of the agricultural enterprise, such as farm size, location and land tenancy (White and Irwin, 1972; Pope and Prescott, 1980; Weiss and Briglauer, 2002; Mishra et al., 2004; Garcia-Arias et al., 2015), and the characteristics of the farmer, such as experience, age, and education (Pope and Prescott, 1980; McNamara and Weiss, 2005; Weiss and Briglauer, 2002; Mishra et al., 2004).

The results of previous studies do not always coincide in the direction of the effects of the different variables on the diversification level. For example, regarding the size of the farm, some authors report a positive relationship between crop diversification and the size of the farm (Pope and Prescott, 1980; Weiss and Briglauer, 2002; Culas, 2006; Ashfaq et al., 2008; Garcia-Arias et al., 2015), while others detect a negative relationship, indicating that an increase in the size of the farm leads to a decrease in diversification (White and Irwin, 1972; Mishra et al., 2004). These contrasting findings have sparked ongoing debate in literature and underscore the complexity of interpreting the role of farm size in shaping diversification strategies. On the one hand, larger farms may encompass heterogeneous site conditions that promote diversification even under profit-maximization objectives (Knoke et al., 2017); on the other hand, diversification in smaller farms may respond to livelihood needs, ensuring the provision of multiple products for household consumption (Pannell et al., 2014). Thus, it is important to understand how these characteristics relate to diversification in the Argentine Pampas.

Argentina's agricultural sector significantly contributes to the national economy, generating over 65 % of export revenues (INDEC, 2022). Furthermore, Argentina ranks among the world's leading exporters of diverse agricultural products, including maize and soybeans (FAOSTAT, 2023). The Pampas alone accounts for 86 % of Argentine

agricultural exports (INDEC, 2022). Despite its importance, limited information is available regarding the factors influencing decisions about agricultural land use in this highly productive region.

Consequently, the objective of this study is to examine the extent to which crop sequence diversification in the core zone of the Argentine Pampas is influenced by multiple objectives, perceived constraints, and the structural and socio-demographic characteristics of farms and their decision-makers. This research seeks to generate empirical evidence on the factors that drive land-use decisions, thus contributing to a better understanding of the determinants of sustainable intensification pathways for the Argentine Pampas.

2. Methods

2.1. Study area

This study focuses on the Arrecifes river basin, which is in the northern part of Buenos Aires province. This basin is a representative area of the Rolling Pampas and part of Argentina's corn belt. It is characterized by high agricultural productivity and considerable heterogeneity in crop sequence strategies, making it an appropriate case for analyzing the factors that influence decision-making regarding crop sequence diversification (Fig. 1). The region is composed of eleven districts, classified into three groups based on their location and characteristics within the basin. Group A, positioned to the north of the basin, encompasses the districts of Pergamino, Rojas, Colón, and Salto. Group B, found in the southern region of the basin, encompasses the districts of Junín, Chacabuco, and General Arenales. Finally, Group C, located to the east of the basin, includes the districts of Baradero, San Pedro, Capitán Sarmiento, and Arrecifes (Fig. 2). Group C exhibits the lowest proportion of class I soil (with no natural restrictions for agriculture). Group B is distinguished by the presence of fine and very fine sands, whereas Group A is primarily marked by coarser sands. Groups A, B, and C have, on average, 45 %, 22 %, and 3 % of Class I soil, respectively (INTA, 1993).

2.2. Sampling methodology

We collected the study data through a survey of farmers within the Arrecifes river basin, conducted between April 2020 and March 2021. A stratified sampling approach was employed. The population list used for sampling was sourced from Argentina's National Service of Health and Food Quality (SENASA). This registry is one of the most comprehensive available in the country and contains information on farms' locations and sizes. The composition of the stratified sample was done based on these two variables, location and farm size. Twelve strata were defined. Each stratum corresponds to one of the quartiles of managed areas within each of the three regions (Fig. 2, Fig. 3). Data collection within each stratum was carried out through a random sampling, where 10 farmers were selected from the population. In cases where contact could not be established with a selected farmer, this one was replaced by another randomly chosen individual. A total of 120 farmers were surveyed. Forty-three of the respondents farmed both owned and leased land and provided separate responses regarding land use under each tenure type, leading to a final sample size of 163 observations. However, the additional 43 observations cannot be considered statistically independent from the 120 other observations, since some farmers contribute two observations. To address this issue, all statistical models accounted for clustering at the farmer level by adjusting standard errors accordingly.

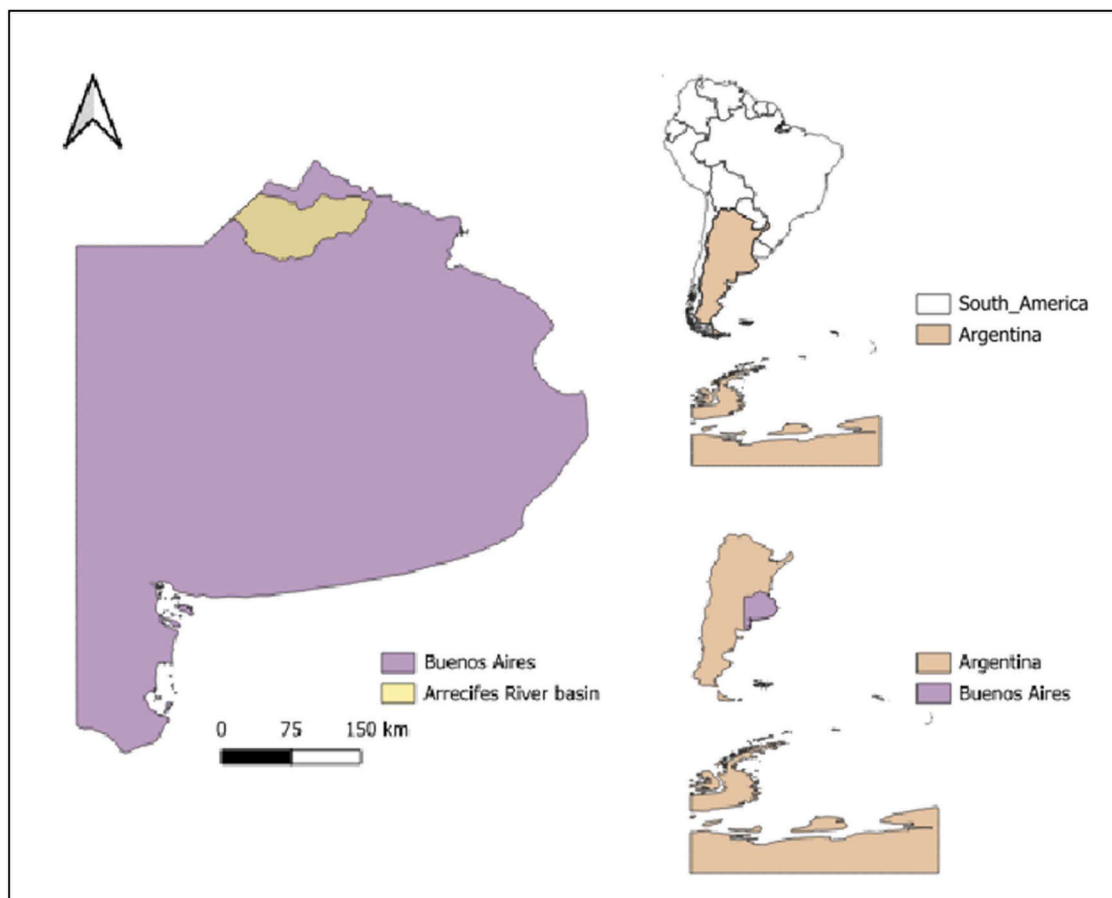


Fig. 1. Location of the Arrecifes River basin.

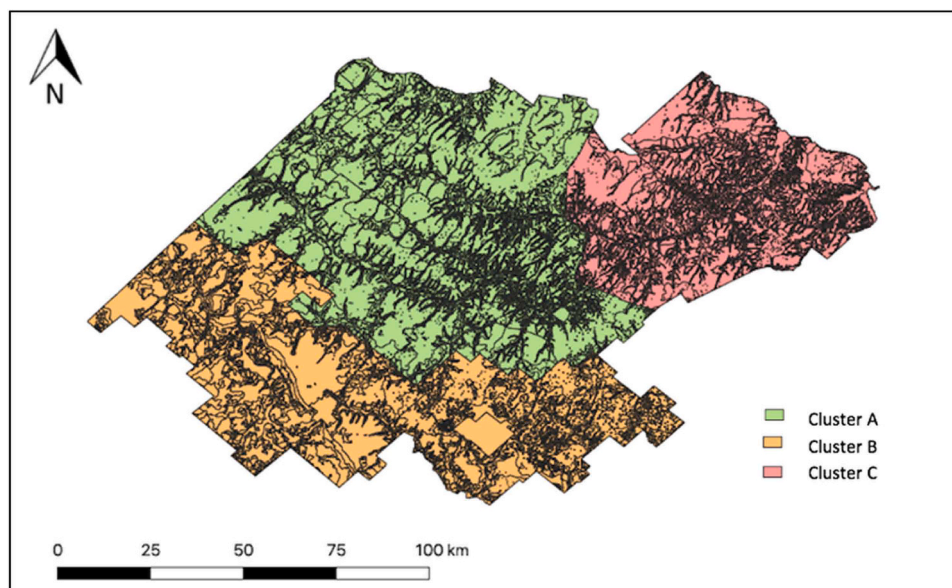


Fig. 2. Groups by district location in the Arrecifes River basin, Buenos Aires state.

2.3. Survey

We designed and implemented a survey instrument to gather demographic and productive information from farmers. The questionnaire collects data on agricultural enterprise characteristics, such as size, land tenure, and location. It also captures attributes of the individual

responsible for land use decisions (age, educational level, years of agriculture experience, and whether agriculture is their main source of income). Additionally, the questionnaire collects information about non-agricultural activities and individuals who are involved in making land use decisions. One of the central questions refers to the cultivated area for different crops during three consecutive agricultural cropping

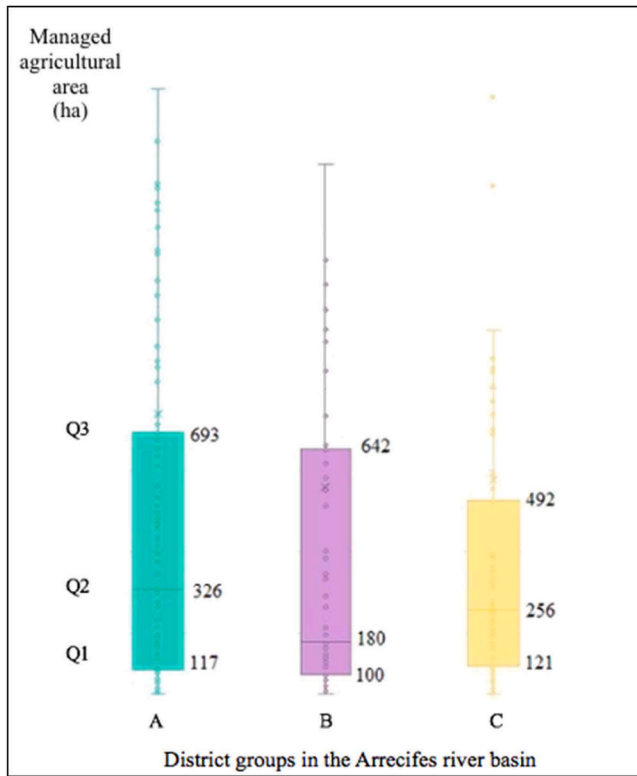


Fig. 3. Quartiles of managed agricultural area (ha) for farmers in the three district groups (Fig. 2) of the Arrecifes river basin.

seasons (2017–2018, 2018–2019, and 2019–2020).¹ Questions about objectives and constraints considered relevant for making agricultural land use decisions were also included in the questionnaire (Annex 1).

2.4. Level of production diversification

We employed the Herfindahl-Hirschman concentration index (HH) to quantify the level of land use diversification (Kumar and Kumar, 2018; Ponce et al., 2015). This index, which ranges from 0 to 10000, is employed regularly in microeconomics to measure the degree of economic activity concentration (Parkin, Loria, 2015). We calculated the HH index based on hectares allocated to each crop during the three surveyed cropping seasons (2017–2018, 2018–2019, and 2019–2020). The computation considered the set of crops reported by farmers in the survey: soybeans, wheat, maize, sunflower, grain sorghum, field pea, barley, and oats, which represent the portfolio of land uses effectively cultivated in the Pampas. Specifically, the sum of squared percentages of land area dedicated to each crop in the three cropping seasons was computed (Eq. 1). The highest HH index value (10000) corresponds to land use with a single crop (i.e., monoculture). As HH index values decrease, levels of production diversification increase.

¹ In general, medium and large farmers, who manage multiple plots, allocate proportions of the land to each crop in each season similar to the participation of these crops in the rotation. In contrast, smaller producers who alternate between various crops may not plant all crops in every season to maintain a minimum scale per crop. This results in the underestimation of the level of diversification in rotations for smaller producers when land use is surveyed for only one crop season. This study addresses this bias by considering three seasons for diversification calculations. Additionally, a question has been included where the producer indicates the most frequent rotation, which is used to further control for this bias.

$$SUP_{ji} = \sum_{t=1}^3 (sup_{jti}) \quad (1)$$

$$Sup_{totali} = \sum_{j=1}^J SUP_{ji} \quad (2)$$

$$HH_i = \sum_{j=1}^J \left(\left(\frac{SUP_{ji}}{Sup_{totali}} \right) * 100 \right)^2 \quad (3)$$

Where:

i is the set of surveyed farmers

j is the set of crops

t is the set of crop years

sup_{jti} is the area allocated to a crop _j in crop year _t by farmer i

SUP_{ji} is total planted area for crop _j for each farmer in the three crop years

SUP_{totali} is total crop area for farmer i

HH_i is the diversification index for each farmer in the sample.

2.5. Probit Statistical Model

We use a Probit statistical model to identify the factors affecting land use decision. This model is commonly used to study choices between alternatives (McFadden, 1973). Its application allows for the examination of the relationship between various factors and farmers' decision-making. In land use choice models, the effects of numerous independent variables have been studied (Rahman, 2008; Mamani, 2016). Analyzing these variables enables the estimation of the probability that certain land use is adopted by a decision-maker or groups of decision-makers within an agricultural enterprise with specific characteristics and motivations (Knowler and Bradshaw, 2007).

The Probit model takes the following form:

$$P(Y = 1|X) = F(X\beta) \quad (4)$$

Where:

X is the vector of explanatory variables, β is the vector of parameters to be estimated, and $F(\cdot)$ is the cumulative normal distribution function, where the estimated values strictly range between zero and one, $0 < F(X\beta) < 1$.

In this study, $P(Y=1)$ corresponds to the probability of choosing a high level of land use diversification ($HH \leq 2500$). The vector of explanatory variables (X_n) encompasses characteristics from the farm characteristics and the decision-maker (Table 1). A cutoff point of $HH = 2500$ was selected as it aligns with a land allocation strategy for farms following a one-third rotation system involving wheat/second soybean, maize, and full-season soybean. Under this scheme, one-third of the land is allocated to each of these three crop types or crop combinations equal proportions of land. In this study, second season soybean (planted after wheat harvest within the same cropping season) was considered as a distinct crop from full-season soybean due to its agronomic differences. This crop sequence is one of the most adopted in the Pampas region (Andrade et al., 2022). This one-third rotation aligns well with sustainable intensification pathways which are agricultural practices that aim to increase productivity while minimizing environmental impact, compared to less diverse sequences, such as soybean/maize, or soybean monoculture (Cano et al., 2023).

Given that some farmers provided responses for both owned and leased land, the sample includes two observations for some individuals. To ensure robust inference, standard errors in the Probit model were adjusted using a cluster-robust variance estimator. This adjustment was performed in R (R Core Team, 2023) using the *clubSandwich* package (Pustejovsky and Tipton, 2022), clustering by individual farmers.

Table 1

Variables select to characterize farms and their decision-makers from the Arrecifes river basin.

	Variables	Unit	Definition
Farm characteristics	Operated land	Hectares	Farm size
	Land tenure	Binary	1 for owned land, 0 for rented
	Group A (North zone of the basin)	Binary	1 for farm located in the North zone, 0 otherwise
	Group B (South zone of the basin)	Binary	1 for farm located in the South, 0 otherwise
	Age	Years	Farm manager's age
	Primary or secondary education	Binary	1 if the highest level of education is primary or secondary, 0 otherwise
Characteristics of the Decision-Maker	Higher education in agricultural sciences	Binary	1 if they have higher education in agricultural sciences, 0 otherwise
	Experience in farming	Years	Farm manager's experience in farming
	Agricultural activity as the main source of income	Binary	1 if agricultural activity is the main source of income, 0 otherwise
	Livestock activity	Binary	1 if they engage in livestock activity in addition to agricultural activity, 0 otherwise
	Decision-making with an advisor	Binary	1 if an advisor is involved in land use decision-making, 0 otherwise
	Decision-making with the family	Binary	1 if the family participates in land use decision-making, 0 otherwise
	Other decision-making	Binary	1 if they make land use decisions differently from the above methods, 0 otherwise)

Note: "Higher education" Higher education refers to tertiary or university-level education.

2.6. Objectives and limitations of decision-makers

In the survey, each farmer was asked to select the most relevant objectives when making decisions about land use. Ten objectives were presented, defined based on objectives identified in land use decision-making literature (Willock et al., 1999; Barbieri, Mahoney, 2009; Hansson et al., 2013), and corroborated in meetings with a team of extension professionals from the National Institute of Agricultural Technology (INTA) of Argentina. Regarding these ten objectives, each farmer was asked to choose the three most relevant ones when making land use decisions. We analyzed the frequencies of different objectives being selected as the top three most important. Additionally, five limitations were presented, and participants were asked to rank them based on their importance when deciding on land use. For analysis, only the limitation ranked first was considered the most restrictive. Separate questions were formulated for decisions regarding land allocation for owned and leased land. We used Odds Ratios (OR) to analyze the relationship between objectives and limitations with diversification. ORs are employed to compare the relative probabilities of the occurrence of the outcome of interest (e.g., diversification), given exposure to the variable of interest (e.g., objective to reduce economic risk) (Szumilas, 2015; Cerda et al., 2013). The odds represent the probability of an event happening divided by the probability of it not happening. An Odds Ratio of 1 indicates no association between the variables. Values below 1 indicate a negative association, while values above 1 indicate a positive association between the variables. To assess each model included a single explanatory variable (a specific objective or constraint) to estimate its individual association with the decision to diversify. Since some

farmers responded on both owned and rented land, the sample included two observations for some individuals. To ensure robust inference, we applied cluster-robust standard errors. This adjustment was performed using the `vcovCR()` function in the `clubSandwich` package in R. P-values were adjusted using the Satterthwaite approximation

3. Results

Table 2 presents farmers' descriptive statistics in the Arrecifes river basin. The reported average operated land was 447 ha. Out of the responses, 60 % corresponds to owned land usage, while the remaining 40 % pertain to leased land usage (Table 2). The average age of the respondents was 47 years old, and the average experience in farming was 24 years. The most common educational level is associated to university education related to agricultural sciences. Eighty-two percent of the sample rely on agricultural activity as their primary source of income, and 50 % of the sample also involves livestock farming. Forty percent of decision-makers make land use decisions independently. (Table 2).

3.1. Land Use

Farmers surveyed indicated that they assign their agricultural land to eight distinct crops: soybeans, wheat, maize, sunflower, grain sorghum, field pea, barley, and oats. Soybeans, maize, and wheat were the prevailing crops. The HH index, determined by the proportion assigned to each crop within the sequence, ranged from a minimum value of 1667 (reflecting maximum diversification) to a maximum value of 10000,

Table 2

Descriptive statistics of surveyed farmers and their farms in the Arrecifes river basin.

Variable	Unit	Mean/ Percentage	Standard Error
Operated land	ha	447	51.47
Land tenure - owned	%	60	0.04
Age	years	46.5	1.09
Experience in farming	years	23.7	1.18
Primary or secondary education	%	30	0.04
Higher education related to agricultural sciences	%	52	0.04
Higher education not related to agricultural sciences	%	18	0.03
Agricultural activity as the main source of income	%	82	0.03
Livestock activity	%	50	0.04
Individual decision-making	%	40	0.04
Decision-making with an advisor	%	24	0.03
Decision-making with the family	%	20	0.03
Other decision-making	%	15	0.03

Note: Total number of observations = 163

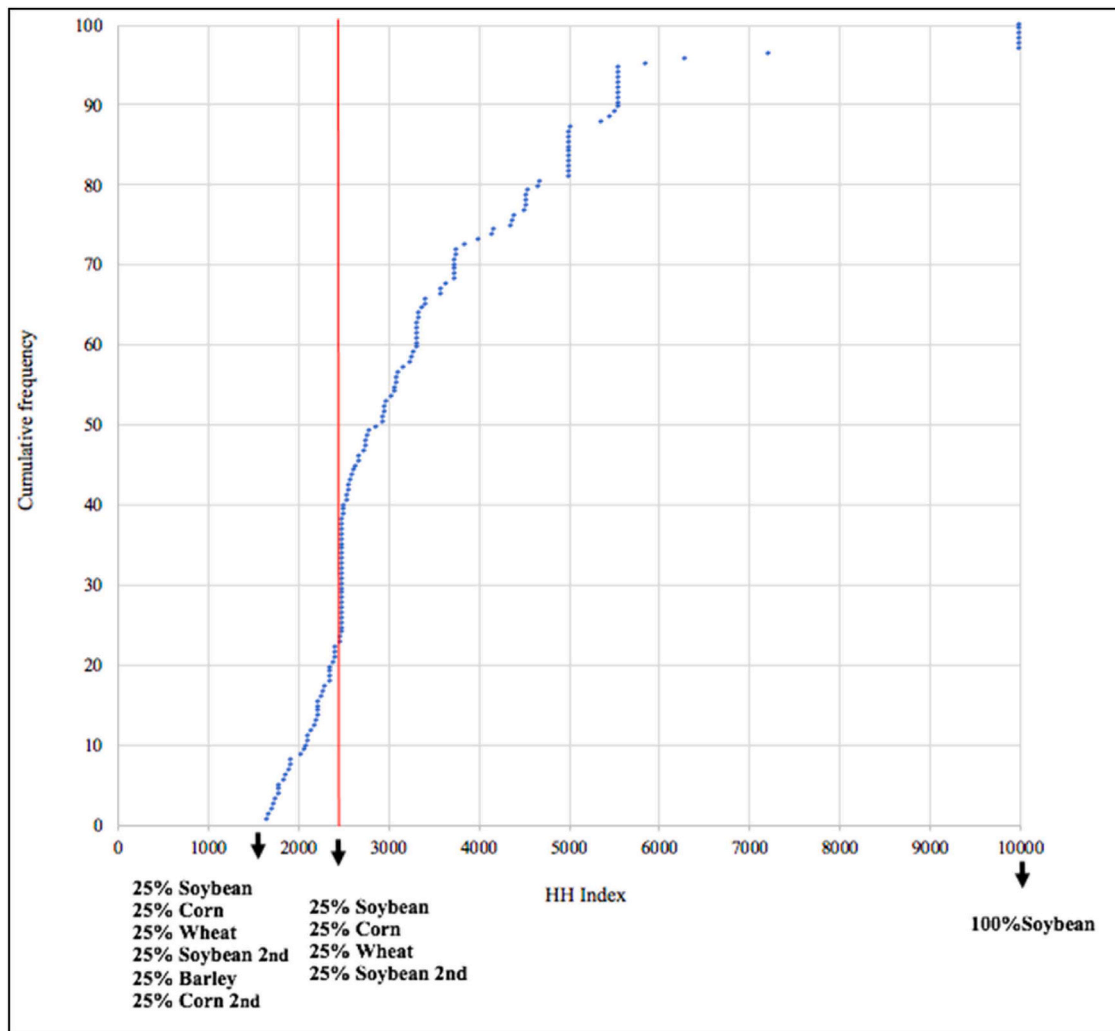


Fig. 4. Cumulative relative frequency of the Herfindahl-Hirschman Concentration Index (HH) for land use in the Arrecifes river basin during the 2017–2018, 2018–2019, and 2019–2020 cropping seasons. Note: Second crop activities were considered as distinct from the first crop activities for the HH calculation.

indicative of soybean monoculture (Fig. 4).

A total of 4 % of the respondents reported the lowest level of diversification, allocating their entire land to soybean monoculture across all three seasons (HH=10000). Approximately, 60 % of the farmers reported less diversified rotations compared to the soybeans-maize- wheat/soybean double crop sequence (HH> 2500).

3.2. Relevant objectives and limitations in land use decisions

Fig. 5 illustrates the frequency of choice for the top three objectives in making land use decisions. Increasing the economic yield of the enterprise, reducing economic risk, and maintaining soil nutrient levels and organic matter were the most frequently chosen objectives. The probability of diversification showed a correlation with the two of the most relevant decision-making objectives. On one hand, the probability of diversification and the selection of the objective "reduce the economic risk level of the enterprise" were significantly negative related (Fig. 5). On the other hand, a significantly positive correlation was found between the level of diversification and the selection of the objective to "maintain a high level of organic matter" (Fig. 5).

Among the constraints that farmers faced when deciding the land use of fields, availability of both land and working capital were the most frequently mentioned (Fig. 6). The probability of diversification was significantly correlated with the limitation of land availability (Fig. 6).

Table 3 shows the estimation results of the Probit model to test for

the relationship between diversification and the farmers' and decision makers characteristics (Eq. 3)

$$P(\text{Diversificación}_i = 1|X) = \beta_0 + \beta_1 OL_i + \beta_2 LT_i + \beta_3 gA_i + \beta_4 gB_i + \beta_5 Age_i + \beta_6 Hs_i + \beta_7 HsR_i + \beta_8 Exp_i + \beta_{10} Ing_i + \beta_{11} Liv_i + \beta_{12} DecA_i + \beta_{13} DecF_i + \beta_{14} DecO_i + u_i \quad (3)$$

Where:

Diversificación *i* is 1 if the HH index is equal to or less than 2500 and 0 if the HH index is greater than 2500; *OL* is operated land; *LT* is land tenure; *gA* is the group of districts located north of the Arrecifes river basin; *gB* is the group of districts located south of the Arrecifes river basin; *Age* is farm manager's age; *Hs* is primary or secondary education, *HsR* is higher education related to agricultural sciences; *Exp* is experience in farming; *Ing* is agricultural activity as the main source of income, *Liv* is livestock activity; *DecA* is Decision-making with advisor, *DecF* is Decision-making with family, *DecO* is Decision-making (other), *u* is the error term. (variables are defined in Table 1)

Note: The model included a correction of the standard error by grouping farmers who work on both owned and rented land. The model error exhibits heteroscedasticity due to the presence of groups of farmers who work on both owned and rented land (120 farmer groups).

Among the farm characteristics, the operated land, land tenure, and belonging to the eastern zone of the basin (Group C) had significant

Objectives	Percentage of farmers	Odds Ratio	t-statistic (Satterthwaite adjusted)	P-value
Increasing the company's economic performance	74%	0.96	-0.1	0.91
Reducing economic risk	58%	0.44	-2.32	0.02**
Maintaining soil nutrient levels	54%	1.24	0.65	0.52
Maintaining a high level of soil organic matter	44%	1.98	2.08	0.04**
Reducing nutrient losses from the system	23%	0.64	-1.05	0.29
Reducing the environmental impact of agrochemical use	17%	0.5	-1.49	0.15
Preventing water erosion	17%	1.87	1.37	0.18
Generating jobs for family members	5%	1	0.005	0.99
Conserving biodiversity	4%	2.32	1	0.35
Generating jobs	4%	1.71	0.59	0.58

Fig. 5. Percentage of farmers who chose each objective within the top three most relevant when deciding the use of agricultural land and Odds Ratios from logistic regressions assessing the association between objectives and diversification. Note: * $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$ **** $p = 0$. Number of observations = 163 (HH > 2500 102 (0); HH ≤ 2500 61(1)).

Constraints	Percentage of farmers	Odds Ratio	t-statistic (Satterthwaite adjusted)	P-value
Availability of land to work	33%	2.21	2.2	0.03**
Availability of working capital for crop expenses	25%	1.07	0.12	0.9
Availability of adequate machinery at the right time	16%	0.81	-0.45	0.65
Availability of workers	15%	0.57	-1.16	0.25
Management capacity	11%	0.62	-1.11	0.26

Fig. 6. Percentage of farmers who chose each constraint as the main one when deciding the use of agricultural land and Odds Ratios from logistic regressions assessing the association between constraint and diversification. Note: * $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$ **** $p = 0$. Number of observations = 163 (HH > 2500 102 (0); HH ≤ 2500 61(1)).

effects on diversification (Table 3). Regarding the decision-makers, characteristics, experience, and livestock activity were statistically significant in the model (Table 3).

Within the farm's characteristics, it has been identified that operated land, land tenure, and location to the east of the basin, in contrast to the southern basin location, showed a significant marginal effect. In this context, it is noteworthy that both land tenure and the eastward location in the basin exerted a more pronounced influence than the amount of operated land, based on the magnitude of the marginal effect (Table 4). Regarding characteristics of decision-maker, experience in farming and livestock activities exhibited a significant marginal effect. Experience had a relatively small negative impact, so engagement in livestock activities emerged as the most relevant factor among the decision-maker's characteristic (Table 4).

4. Discussion

In the Argentine Pampas, the decision to diversify cropping sequences is related to the characteristics of the agricultural enterprises and their decision-makers. Firstly, the characteristics of the agricultural enterprise that influence the decision to diversify cropping sequences are the operated land, land tenure, and location within the Arrecifes

river basin. Secondly, engaging in livestock activity and the experience of the farmer were the most prominent characteristics describing the decision-makers profile that influenced the willingness to diversify cropping sequences (Table 3). The probability of diversification is lower for farmers with "reducing the level of economic risk of the enterprise" as one of the main objectives. On the other hand, the probability of diversification is higher for farmers with "maintaining a high level of organic matter" as one of the main objectives (Fig. 5). Finally, the probability of diversification was significantly positively correlated with the limitation of land availability for cultivation (Fig. 6). The following paragraphs contain a discussion regarding the factors affecting land use.

4.1. Operated land of the farm

Diversification was positively correlated with the operated land (size) of the farm. The farm size was considered as the hectares planted during three consecutive cropping seasons (2017–2018, 2018–2019, and 2019–2020). For each additional hectare in the farm, the decision-maker has a 0.02 % higher probability of opting for more diversified land uses (Table 3). Our results align with findings from other studies (Pope and Prescott, 1980; Weiss and Briglauer, 2002; Culas, 2006; Ashfaq et al., 2008).

Table 3

Estimated coefficients of the Probit model for the possibility of choosing a higher level of diversification as the response variable.

	Variables	Coefficients	Robust Standard Error
Farm characteristics	Operated land	0.0006***	0.0002
	Land tenure- owned	0.62**	0.31
	Group A (north zone of the basin)	0.22	0.27
	Group C (south zone of the basin)	0.50**	0.30
	Age	0.02	0.017
Characteristics of the Decision-Maker	Primary or secondary education	0.31	0.42
	Higher education in agricultural sciences	0.07	0.36
	Experience in farming	-0.03**	0.015
	Agricultural activity as the main source of income	-0.22	0.33
	Livestock activity	0.56**	0.23
	Decision-making with an advisor	-0.44	0.35
	Decision-making with the family	-0.16	0.36
	Other decision-making	-0.09	0.34
	Constant	-1.52	0.72

Note: *p < 0.1 **p < 0.05 ***p < 0.01

Number of observations= 163

Wald Chi² (13) = 31.92Prob>Chi²= 0.0025Pseudo R²= 0.13

Log pseudolikelihood= -93.64

Standard Error Adjusted to 120 decision-makers with owned and leased land

The positive relationship between diversification and the size of the farm was explained in different ways. Pope and Prescott (1980) suggest that greater diversification in larger farms is consistent with risk theory, as companies diversify to spread risk. Culas (2006) explains the positive relationship between diversification and the size of the farm due to the lack of economies of scale in Norwegian agriculture. Ashfaq et al. (2008) explain it by the fact that larger farmers have more land resources to allocate to different crops and can dedicate more attention to agricultural activity compared to other activities outside the farm. In the Argentine Pampas region, the positive relationship between diversification and the size of the farm can be explained similarly to what Ashfaq et al. proposed in 2008. This relationship is based on the more efficient utilization of land and time resources with a larger land area for cultivation, which stimulates diversification.

4.2. Land Tenure

Diversification was positively correlated with land tenure. The decision-maker has an 81 % higher probability of opting for less diversified land uses when the land is leased. In the Pampas region, Arora et al. (2015) observed that farmers place more emphasis on short-term economic outcomes, motivated by the initial cost of leasing, when land is managed under a lease arrangement, even when there is continuity in lot management. In such a context, the negative correlation between the proportion of leased land and the diversification of cropping sequences seems a rational decision. The decision-makers would

tend to choose simplified sequences with a high prevalence of soybeans, as it is the crop with the highest ratio grass margin/direct costs (Lanteri, 2009; Pengue, 2009).

4.3. Location of the farm

Diversification of cropping sequences was associated with the location of the farm in the Arrecifes river basin. Farms located in the eastern basin (Group C) had a 17 % higher probability of opting for diversified cropping sequences compared to those located in the southern basin (Group B). Group C, located in the eastern basin, has the lowest proportion of Class I soil, which is the best soil class for agriculture. Therefore, it is likely that these fields have livestock activity complementing agriculture. Engaging in livestock activity involves the incorporation of pastures, winter and summer forage crops, and a higher proportion of maize, leading to more diversified cropping sequences. These results are like those found by Cutforth et al. (2001), Roesch-McNally et al. (2018), and Wang et al. (2021), where marginal lands were positively associated with diversification.

4.4. Livestock activity

When the decision-maker engages in livestock activity in addition to agriculture, there is a 20 % higher probability of opting for more diversified cropping sequences. "Crop rotation is planned based on livestock", said a farmer who produces both crops and livestock (Field

Table 4

Marginal effects for the farm characteristics and the decision-maker.

	Variables	Marginal effect	Standard Error
Farm characteristics	Operated land	0.0002***	0.000
	Land tenure - owned	0.20**	0.09
	Group A (north zone of the basin)	0.07 **	0.08
	Group C (south zone of the basin)	0.17**	0.09
	Age	0.01 ***	0.006
Characteristics of the Decision-Maker	Primary or secondary education	0.1	0.13
	Higher education in agricultural sciences	0.02	0.11
	Experience in farming	-0.01 ***	0.005
	Agricultural activity as the main source of income	-0.07	0.10
	Livestock activity	0.18**	0.07
	Decision-making with an Advisor	-0.13	0.11
	Decision-making with the Family	-0.05	0.11
	Other Decision-making	-0.03	0.11

Nota: *p < 0.1 **p < 0.05 ***p < 0.01

Day of the *Jornada a Campo del IPCVA*, 2022). The fact that cropping sequences are planned with a focus on livestock entails the incorporation of pastures, winter and summer forage crops, and a higher proportion of maize in the cropping sequence, resulting in more diverse sequences. These results coincide with other studies that have demonstrated a positive relationship between the incorporation of livestock into agriculture and the diversification of crop sequences (Cutforth et al., 2001; Russelle et al., 2007; Roesch-McNally et al., 2018).

This behavior reflects the concept of economies of scope, where farmers achieve cost efficiencies by producing complementary outputs—crops and livestock—that share resources such as land, labor, and machinery. For instance, incorporating forage crops or pastures into rotations not only supports livestock feed requirements but also enhances soil health and reduces input costs for subsequent crops. This contrasts with economies of scale, which focus on cost reductions achieved through the expansion of a single enterprise (e.g., large-scale monocropping systems).

4.5. Experience in farming

Our results show a negative relationship between diversification and the decision-maker's experience. For each additional year of experience in farming that the decision-maker has, the probability of diversification is 1 % lower. The experienced farmer is likely less willing to try different crops, as they have found a stable and simpler system for decision-making. This negative relationship between diversification and experience might also be related to changing economic structures over time. Ochoa et al. (2023) observed that as farmers gain experience over years, many increasingly engage in off-farm activities that generate additional income, which subsequently leads to simplified farming systems that require less management. However, a study conducted by Ashfaq in 2008 in Pakistan found a non-significant and positive relationship between diversification and the decision-maker's experience. It is important to note that the systems analyzed in that study differ from those examined in this work, as they involved crops such as rice, sugarcane, and wheat. Additionally, in a study carried out by Culas in 2006 in Norway, a positive relationship between diversification and the decision-makers' experience was also found. However, the systems studied in that case differ from those analyzed in this work, as they include not only cereal crops but also vegetables, fruits, and several livestock activities. Furthermore, the farms analyzed in the current study are smaller, with an average size of 25 ha.

4.6. Farmer's objectives in deciding land use

The probability of diversifying cropping sequences is positively related to the selection of the objective to "maintain a high level of organic matter." This result is consistent with the fact that simplified sequences with a high proportion of soybeans usually have a lower contribution and quality of residues compared to more diversified sequences, and therefore, the storage of organic carbon in the soil is negatively affected (Franzuebbers et al. 1998; Studdert, Echeverria, 2000; Wright and Hons, 2004). It also supports the findings of Gosling et al. (2020), who observed that farmers prioritizing ecological goals—such as soil health and long-term sustainability—tend to implement more diversified production systems compared to those driven mainly by short-term economic considerations.

Moreover, the likelihood of diversifying cropping sequences is inversely related to the selection of the objective to "reduce the level of economic risk of the enterprise." This counterintuitive relationship contrasts with expectations derived from portfolio diversification theory, which identifies the role of diversification as a risk reduction strategy. This finding also contrasts with the concept of crop biodiversity as a form of natural insurance, suggesting that greater crop diversity can buffer against environmental risks and market fluctuations (Baumgärtner and Quaas, 2010; Di Falco and Perrings, 2005). However,

several factors may explain why more risk-averse farmers prefer less diversified crop sequences.

In the Pampas, less diversified crop sequences consistently include soybeans as the predominant crop, likely perceived by farmers as a low-risk activity due to lower demand for working capital per hectare (Cabrini and Calcaterra, 2016). Incorporating various crops into a sequence involves planting smaller-market crops (such as field pea, barley, and oats), which are perceived as riskier alternatives due to limited knowledge in crop management technology, fewer buyers, and greater price volatility. Furthermore, farmers with access to off-farm income may deliberately simplify their agricultural operations to reduce management complexity while maintaining a stable, if limited, source of production (Ochoa et al., 2019). In doing so, they diversify their overall income portfolio rather than their cropping systems as a strategy to manage household-level risk.

These findings challenge traditional assumptions about diversification as a universal risk management strategy. Instead, they highlight how contextual factors such as market dynamics, input costs, and access to alternative income sources shape farmer decision-making in ways that deviate from theoretical expectations. Addressing these barriers will require targeted interventions such as training programs on alternative crop management practices or market incentives for less mainstream crops.

4.7. Limitations faced by the farmer in land use decisions

The probability of diversification was positively related to the limitation of land availability for cultivation. Farmers who consider land as the most limiting resource for production are more likely to opt for land uses that generate higher results per unit of area rather than per invested capital. This behavior is consistent with the resource allocation optimization under constraints described by Bikhchandani et al. (2013), where decision-makers adapt their strategies based on their most binding constraints. More diversified sequences are reasonable choices in this situation (Andrade et al., 2022).

4.8. Limitations and future research opportunities

One potential limitation of this study is the lack of a direct measure of off-farm income, which has been shown to influence land-use decisions and crop diversification (Murphy, 2001; Pfeiffer et al., 2009; Araujo et al., 2019; Ahmadzai, 2020; Vasco et al., 2020; Caulfield et al., 2021). Ochoa et al. (2019) found that off-farm income can lead to the homogenization of cropping systems in small-scale farming contexts, as farmers with significant non-agricultural income tend to simplify their agricultural practices. Although our survey did not explicitly capture detailed data on non-agricultural income sources, we included two variables that serve as proxies: (1) whether crop production was the household's primary income source, and (2) whether the household was also engaged in livestock activities. These indicators provide partial insight into the household's income diversification strategy, yet may not fully account for the effects of off-farm income. Future research should consider including more comprehensive measures of income diversification to better assess its impact on crop diversification decisions.

5. Conclusions

Our study examined how different farm characteristics, and the profiles of decision-makers influence the adoption of diversified cropping sequences. The findings show that diversification is positively associated with land size, land tenure, and engagement in livestock activity. In addition, farms located in the eastern basin were more likely to implement diversified crop rotations than those in the southern basin. In contrast, more years of experience with the decision-maker were negatively related to crop diversification.

We also explored the objectives and constraints farmers consider

when making land-use decisions and how these factors relate to diversification levels. Farmers who prioritized maintaining soil organic matter were more likely to diversify, whereas those aiming to reduce economic risk tended to adopt less diverse systems. Among the limitations, land availability emerged as a key factor positively linked with higher levels of diversification.

These insights are valuable for understanding patterns of land use and can inform public policies that aim to support sustainable intensification. Given the broader benefits of crop diversification observed in the region—such as improved soil productivity, enhanced ecosystem services, and better nutrient balance—it becomes crucial to consider policy measures that encourage these practices. In this context, some provinces in Argentina have started implementing tax incentives for landowners who actively farm their land, which could indirectly support more diverse cropping systems. Additionally, public investment in research and extension services focused on the management and commercialization of alternative crops can play a key role in reducing the uncertainties associated with their adoption.

Finally, the country's export tax structure has historically influenced crop choices. Higher logistics costs for less commonly traded crops result in a proportionally larger price impact from export taxes. Any discussion on modifying these tax policies should consider their potential consequences on the diversification of agricultural systems.

CRedit authorship contribution statement

Priscila B. Cano: Writing – review & editing, Writing – original draft, Investigation, Formal analysis, Data curation, Conceptualization. **Silvina M. Cabrini:** Writing – review & editing, Supervision, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Santiago L. Poggio:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Investigation, Formal analysis, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no competing interest.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.landusepol.2025.107778](https://doi.org/10.1016/j.landusepol.2025.107778).

Data availability

Data will be made available on request.

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